

Homework 2 - Local Search

100pts

Due Nov 15 at 11pm

Goal: To learn more about local search which has very low memory usage and can be quite successful for many problems; and to gain further experience with programming and reporting research results.

Task Overview: The task is to solve the N -queen problem with different local search algorithms. N -queen problem is finding the placement of N queens on an $N \times N$ board such that none of the queens one attacks another. You will implement basic hill climbing, hill climbing with random restarts and stochastic hill climbing and then run simulation experiments with each method.

In more detail, **answer/complete the following and fill the appropriate rows in Table 1.**

a) and b) (30+30 points) Implement basic **Hill Climbing** and **Hill Climbing with Random Restart**. Run 100 simulations/experiments with different initial solutions and fill in Table 1 with the number of successes, and the time it takes to find the solution on average for $N=20$, using basic **Hill Climbing** and **Hill Climbing with Random Restarts** with two values for the number of restarts ($k=10, 100$).

c) (30 points) Implement **Stochastic Hill Climbing** and fill the results of 100 experiments to the corresponding row of Table 1.

d) (10 points) Based on your observations, which of the three hill climbing variations (basic, stochastic or random restart) performed well - just summarize in 1-2 sentences.

Submission Guidelines

- Jupyter Notebook: Must include all code cells and outputs. (The notebook will not be re-run during grading.) Must be fully commented before each code cell (also include name, date and relevant info such as extent of AI usage on top of your notebook).
- A pdf document with your filled table (see the next page). You can use Overleaf for editing a given LaTeX template¹. Alternatively, you can prepare your table in MS Word or another tool and export it as a pdf. *Handwritten, hand-filled tables will lose 10 pts.*
- File Naming:
 - CS404-HW2-YourName.ipynb
 - CS404-HW2-YourName.pdf
- Late Policy: –10 points per day, up to 3 days.

Table 1: Simulation results. You will submit a Pdf of this page.

	N=20	
	Percentage of success in 100 runs	Elapsed time to complete experiment (secs)
a) Basic Hill Climbing		
b) Random Restart with k=10		
b) Random Restart with k=100		
c) Stochastic Hill climbing		
d) Enter your short summary		

See the next page for a hint.....

¹<https://www.overleaf.com/read>

```
def calculateNumberOfConflicts(board):  
    """  
    returns number of conflicts  
    since each queen is in one column, no vertical conflicts  
    """  
    numberOfConflicts = 0 # initialize as 0  
  
    for i in range(0, len(board)):  
        for j in range(i + 1, len(board)):  
            if board[i] == board[j]:  
                numberOfConflicts += 1 # increment if horizontal conflict  
            elif abs(i - j) == abs(board[i] - board[j]):  
                numberOfConflicts += 1 # increment if diagonal conflict  
  
    return numberOfConflicts
```